Electronic Markets

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Algorithmic Investments

Algorithmic investments: the trading of financial products guided and/or executed by mathematical models and computer algorithms, typically in electronic markets.

- Financial contracts and empirical facts.
- Mathematical models and analysis.
- Computer algorithms and implementation.

Financial Contracts

Many types of financial contracts (products, assets, etc) are traded in electronic markets today.

- Shares/Company stocks: claims of ownership on corporations.
  - Have to be listed, via initial public offering (IPO), to be traded in electronic markets.
  - The share holders have the right to receive an equal share of the corporation's profits and to intervene (via the right to vote) in the corporate decision processes $\Rightarrow$ Common stock.
- Bonds: contracts by which the corporation/governments/... commit to pay the holder a regular income (interest/coupon), but gives them no decision rights.

Hybrid stock and bonds products: preferred stock, convertible bonds, etc.

Financial Contracts (continued)

Financial contracts can also be categorized according to their characteristics.

- Money market accounts, savings deposits, treasury bills, etc.
- Equities: shares, preferred stock, etc.
- Bonds.
- Foreign Exchange (FX), cryptocurrencies, commodities, real estate, property, etc.
- Mutual funds/hedge funds, exchange-traded funds (ETFs), financial derivatives (e.g., futures, options, swaps), etc.

Market Participants

Asset suppliers: corporations, governments, financial management companies, etc.

Traders.

- Proprietary traders (informed): who trade on a (sometimes real, sometimes illusory) trading advantage, e.g., hedge funds, day traders, etc, typically using short-term strategies.
- Fundamental (noise or liquidity) traders: who have a direct use for the assets being traded regardless of market events, e.g., to share the growth, risk hedging, etc, typically using medium- and long-term strategies.
- Market makers: professional traders who profit from facilitating exchange and exploit their skills in executing trades, typically using high frequency trading strategies.

Trading in Electronic Markets

Electronic markets: a way for people to signal their willingness to trade, and a matching engine to match those wanting to buy and those wanting to sell.

Different types of orders.

- Market orders (MOs): aggressive orders that seek to execute a trade (buy or sell) immediately at the best price available.
- Limit orders (LOs): passive orders that indicate the desire to buy or sell at a given price up to a certain, maximum quantity of shares. LOs may not result in an immediate trade, but may instead wait for a while to match with a new order or be withdrawn (cancelled).

Exchange Structures

Orders are managed by a matching engine and a limit order book (LOB).

Most electronic markets use a price-time priority. When a market buy order comes,

- it first match the sell limit orders with the lowest sell price; the earlier order, the higher priority;
- If there are quantities remained, the order goes deeper to the second-best sell-price limit orders, and so on.

Alternative exchange structures.

- The Prorata rules: MOs are matched against the posted LOs available at the best price in proportion to the quantities posted.
- An initial and/or a closing auction, or auction after a market trading halt.
- Lit (e.g., open order book) versus dark pool markets.

Other Types of Orders

Fragmented markets: an asset can be traded in different markets.

- Non-routable: avoid default re-routing to other exchanges;
- Hidden: orders that do not display their quantity;
- Immediate-or-cancel: orders that execute as much as possible at the best price and the rest are cancelled.
- ...

Exchange fees: may depend on the types of orders.

- Maker-taker: a trader whose posted LO is filled by the MO will pay much lower trading fee, or even receive a payment.
- Taker-maker: inverse the above.

The Limit Order Book

LOB events.

- Addition of LO to LOB: an incoming LO joins the LOB at the last of the corresponding price level.
- MO walks the LOB or is re-routed.

Tick: the difference between one price level and the next on the discrete price grid of LOB.

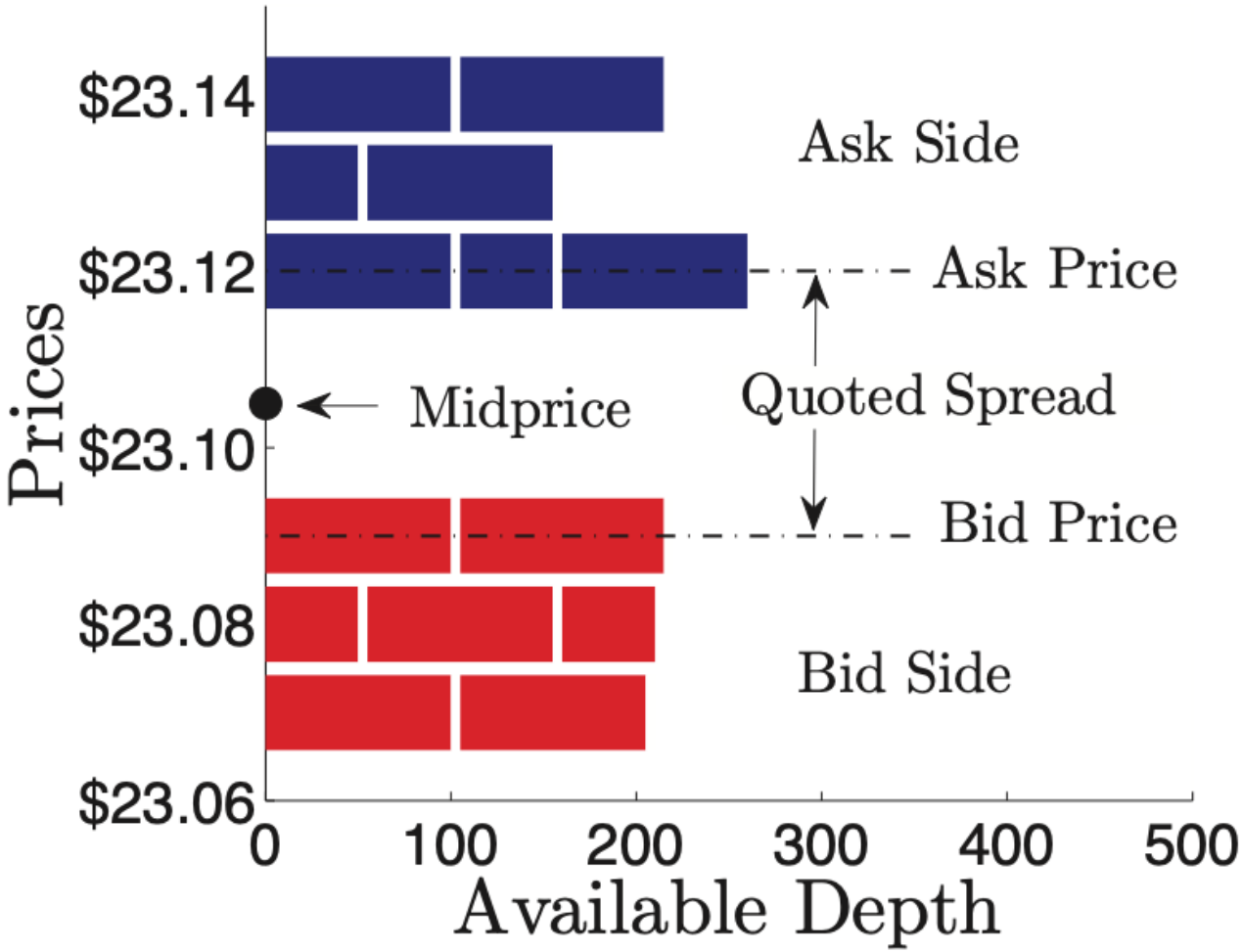
- 1 cent for all stocks with a price above 1 dollar in US.

The bid and ask prices.

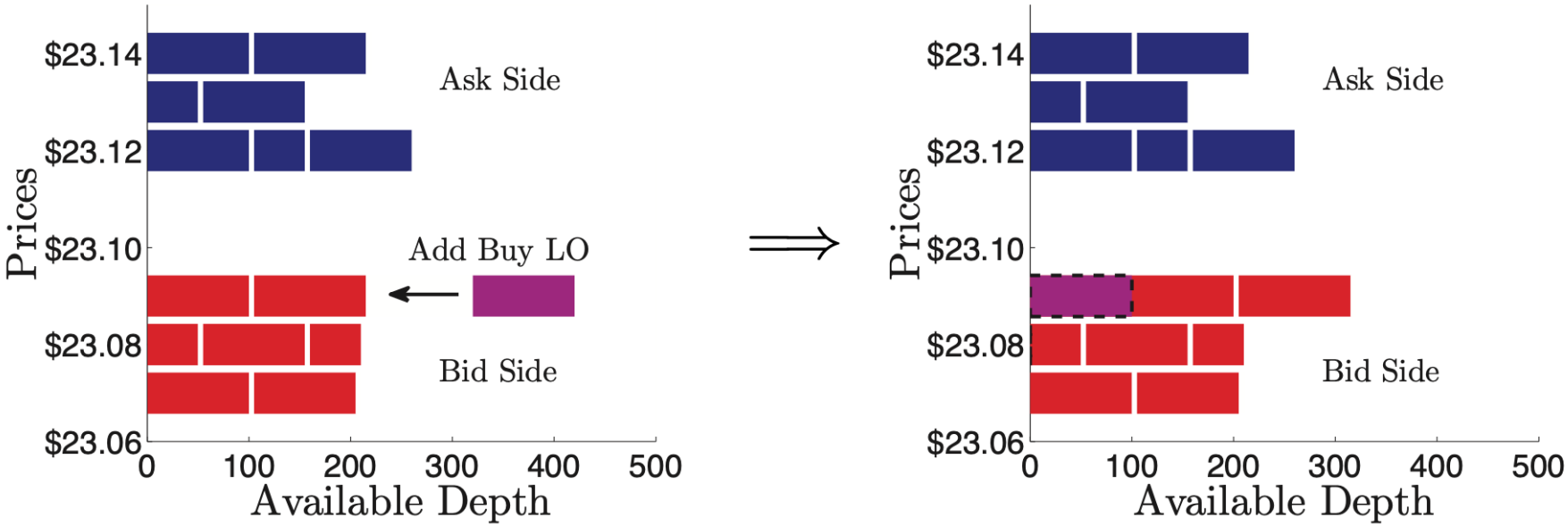
- The bid: the best buy price.
- The ask: the best sell price.

The bid-ask spread: ask minus bid. Mid price: the average of the ask and bid.

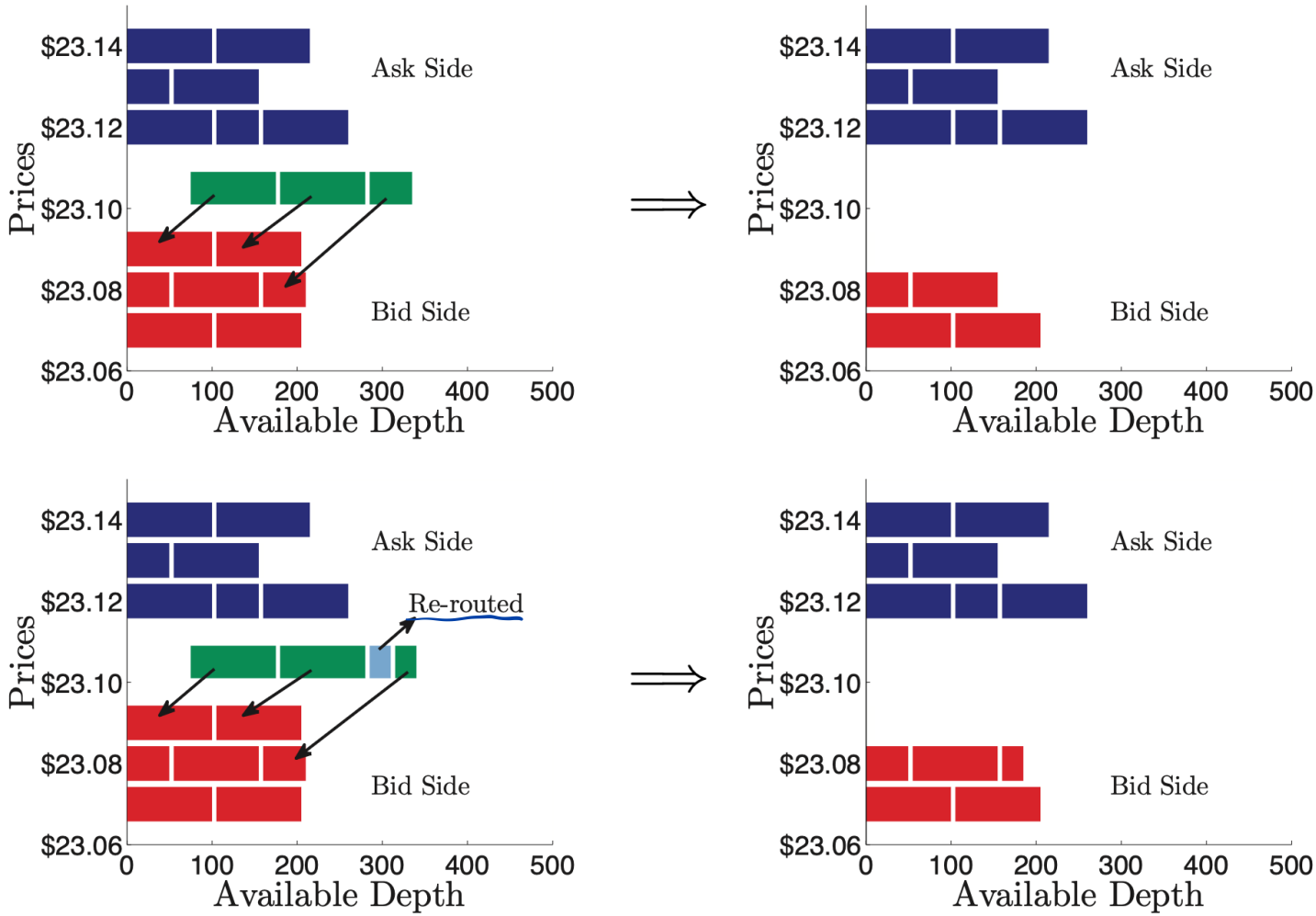
A Snapshot of a Limit Order Book



A Snapshot of a Limit Order Book (continued)



A Snapshot of a Limit Order Book (continued)



The Microprice

A proxy for the asset's cost-free price.

这里没有是反的，描述的是冲击趋势。

$$\text{Microprice}_t = \frac{V_t^b}{V_t^b + V_t^a} P_t^a + \frac{V_t^a}{V_t^b + V_t^a} P_t^b.$$

(接下来变化的方向)

量多的一边更可能成交。

价格倾向于走一边跑。

The Grossman-Miller Market Making Model

Intuition: the market makers (MMs) bear inventory risk, and will require compensation for providing liquidity.

The model setting.

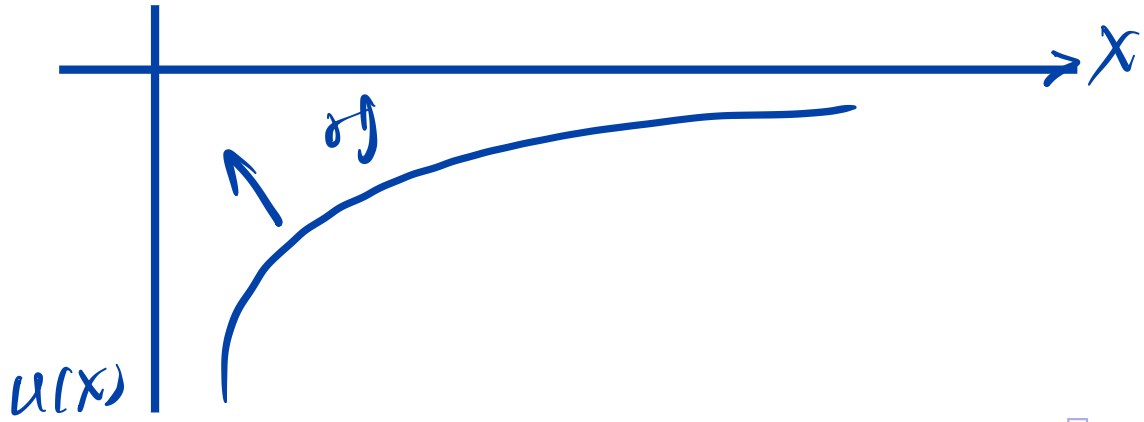
- There are three dates $t = 1, 2, 3$, and n identical MMs.
- At $t = 1$, a liquidity trader (LT1) will sell i units of the asset.
- At $t = 2$, another liquidity trader (LT2) will purchase i units of the asset.
- All agents start with an initial cash W_0, MMs hold no assets, LT1 holds i units, and $LT2$ holds $-i$ units.
- No trading costs, and no direct costs of holding inventory.
- The asset will have a cash value $S_3 = \mu + \epsilon_2 + \epsilon_3$ at $t = 3$ with μ being a constant, and ϵ_2 and ϵ_3 are independent normal random variables (with mean 0 and variance σ^2) announced between $t = 1, 2$ and $t = 2, 3$, respectively.

The Grossman-Miller Market Making Model (continued)

The agents are to make decisions to maximize their expected utility:

$$U(X) = -e^{-\gamma X}, \qquad \gamma > 0.$$

Outcome: the optimal trading behaviors and equilibrium prices.



The Decisions at $t = 2$

Suppose the agents leave $t = 1, 2$ with asset holdings $q^j_{1,2}$ and cash $X^j_{1,2}$, $j = \{MM, LT1, LT2\}$.

Then we have that

$$X^j_3 = X^j_2 + q^j_2 S_3, \; X^j_2 + q^j_2 S_2 = X^j_1 + q^j_1 S_2.$$

The expected utility at this stage:

$$\mathbb{E}[U(X^j_3)|\epsilon_2] = -\exp\Big\{-\gamma\big(X^j_2 + q^j_2 \mathbb{E}[S_3|\epsilon_2]\big) + \frac{1}{2}\gamma^2 (q^j_2)^2 \sigma^2\Big\}.$$

The optimal solution:

$$q^{j,*}_2 = \frac{\mathbb{E}[S_3|\epsilon_2] - S_2}{\gamma\sigma^2},$$

identical for all the agents.

The Supply-Demand Equilibrium at $t = 2$

The asset loadings leaving $t = 2$ are identical. Then

$$\sum_{j=1}^{n} q_2^{j,*} + q_2^{LT1,*} + q_2^{LT2,*} = (n+2)\frac{\mathbb{E}[S_3|\epsilon_2] - S_2}{\gamma\sigma^2} = i + (-i) = 0.$$

Then, in equilibrium,

$$S_2 = \mathbb{E}[S_3|\epsilon_2] = \mu + \epsilon_2,$$
$$q_2^{j,*} = 0, \qquad j = 1, 2, \ldots, n, LT1, LT2.$$

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The Decisions at $t = 1$

From the previous analysis, we have that

$$X_2^j = q_1^j S_2, \; X_1^j + q_1^j S_1 = X_0^j + q_0^j S_1.$$

Repeating the derivations for $t = 2$, we get

$$q_1^{j,*} = \text{argmax}_{q_1^j} \mathbb{E}[U(X_2)] = \frac{\mathbb{E}[S_2] - S_1}{\gamma\sigma^2}.$$

In equilibrium,

$$(n+1)q_1^{j,*} = (n+1)\frac{\mathbb{E}[S_2] - S_1}{\gamma\sigma^2} = i.$$

This implies that

$$S_1 = \mu - \gamma\sigma^2 \frac{i}{n+1}, \; q_1^{j,*} = \frac{i}{n+1}.$$

The equilibrium price is lower (higher) than the efficient price μ if $i > 0$ ($i < 0$).

Trading Costs

Assumptions: the trading cost per share is η, regardless of buying or selling; $i > 0$.

The optimal asset holding leaving $t = 2$:

$$q_2^{j,*} = \frac{\mathbb{E}[S_3 - \eta|\epsilon_2] - (S_2 - \eta)}{\gamma\sigma^2} = \frac{\mathbb{E}[S_3|\epsilon_2] - S_2}{\gamma\sigma^2}, \qquad j = MM, LT1,$$

$$q_2^{j,*} = \frac{\mathbb{E}[S_3 + \eta|\epsilon_2] - (S_2 + \eta)}{\gamma\sigma^2} = \frac{\mathbb{E}[S_3|\epsilon_2] - S_2}{\gamma\sigma^2}, \qquad j = LT2,$$

and then $S_2 = \mathbb{E}[S_3|\epsilon_2]$; the same as before.

Trading Costs (continued)

The asset loadings leaving $t = 1$:

$$q_1^{LT1} = \frac{\mathbb{E}[S_2 - \eta] - (S_1 - \eta)}{\gamma\sigma^2} = \frac{\mathbb{E}[S_2] - S_1}{\gamma\sigma^2},$$
$$q_1^{MM} = \frac{\mathbb{E}[S_2 - \eta] - (S_1 + \eta)}{\gamma\sigma^2} = \frac{\mathbb{E}[S_2] - S_1 - 2\eta}{\gamma\sigma^2}.$$

The market equilibrium condition:

$$nq_1^{MM} + q_1^{LT1} = i \Rightarrow S_1 = \mu - \gamma\sigma^2 \frac{i}{n+1} - 2\frac{n}{n+1}\eta,$$

Moreover,

$$q_1^{LT1,*} = \frac{i}{n+1} + 2\frac{n}{n+1}\frac{\eta}{\gamma\sigma^2}.$$

Price Impact and Market Liquidity

Liquidity: how easy (speed and cost) it is to buy or sell an asset.

Suppose we write:

$$S_1 = \mu + \lambda q_1^{L^T 1},$$

then λ measures the price impact.

Under the Grossman-Miller model,

$$\lambda = -\frac{1}{n}\gamma\sigma^2.$$

Market Making using Limit Orders

The model setting.

- The LOB distribution: once an MO comes, the probabilities that it walks above the depth x are as follows:

$$\begin{cases} P_-(x) = e^{-\kappa^- x}, & \text{buy LO with size } x, \\ P_+(x) = e^{-\kappa^+ x}, & \text{sell LO with size } x. \end{cases}$$

- An MO order is a sell (buy) with probability p^- (p^+).

To maximize the expected profit per trade:

$$\max_{\delta^+,\delta^-} \mathbb{E}[\Pi(\delta^+,\delta^-)] = \max_{\delta^+,\delta^-} \{p^+ e^{-\kappa^+\delta^+}\delta^+ + p^- e^{-\kappa^-\delta^-}\delta^-\}.$$

The optimal solution:

$$\delta^{\pm,*} = 1/\kappa^{\pm}.$$

Trading on an Informational Advantage (Kyle, 1985)

The model setting.

- The market for an asset only opens at one point of time. It is traded at price S (endogenously determined), and after trading, its cash value becomes $v \sim \mathcal{N}(\mu, \sigma^2)$.
- There are three types of traders.
  - Informed (insider) trader who knows the realized value of v, based on which determines the traded amount $x(v)$.
  - Liquidity traders with an aggregate demand for the asset $u \sim \mathcal{N}(0, \sigma_u^2)$.
  - Market makers who are risk-neutral and set the asset price as

$$S(x+u) = \mathbb{E}[v|x+u].$$

- To determine a pricing function $S(\cdot)$ and the insider trading $x(\cdot)$.

Trading on an Informational Advantage (continued)

Hypothesis on the pricing function:

$$S(x+u) = \mu + \lambda(x+u),$$

for constants μ and λ.

The insider's problem:

$$\max_x \mathbb{E}\big[x(v - S(x+v))|v\big] \Rightarrow x^*(v) = \beta(v-\mu),$$

where $\beta = 1/(2\lambda)$.

Let $y = \beta(v-\mu) + u$.

$$\begin{aligned} S &= \mathbb{E}[v|x+u] = \mathbb{E}[v|\beta(v-\mu)+u] \\ &= \mathbb{E}[v|y] = \mathbb{E}[v] + \frac{\mathrm{Cov}[v,y]}{\mathrm{Var}[y]}(y - \mathbb{E}[y]) = \mu + \frac{\beta\sigma^2}{\beta^2\sigma^2 + \sigma_u^2} y. \end{aligned}$$

The Equilibrium

Equilibrium:

$$\lambda = \frac{\beta\sigma^2}{\beta^2\sigma^2 + \sigma_u^2} = \frac{1}{2\beta}.$$

Then $\beta^2\sigma^2 = \sigma_u^2$, and hence

$$\lambda = \frac{\sigma}{2\sigma_u}.$$

Mean-Variance Portfolio Selection

The mean-variance portfolio selection problem is formulated as:

$$\max_x \mu^\top x - \frac{1}{2}\gamma x^\top \Sigma x,$$

where γ is the risk-aversion coefficient.

The optimal solution is:

$$x^* = \frac{1}{\gamma}\Sigma^{-1}\mu.$$

Discret-time versus Continuous-time Formulations

This course focuses on the continuous-time formulation, mainly due to tractability and interpretability.

The mathematical tools include:

- Stochastic control and optimal stopping.
- Reinforcement learning.

Applications include: portfolio selection, execution, market making, pairs trading, etc.